Equations of Parallel & Perpendicular Lines — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Parallel and Perpendicular Slopes

Watch Me — Worked Example

$y = -\frac{4}{5}x + 2$ has slope $-\frac{4}{5}$. Parallel slope: $-\frac{4}{5}$. Perpendicular slope: flip and change the sign, $\frac{5}{4}$.

Give the parallel slope and the perpendicular slope.

1. $y = 3x - 7$

$3;\ -\frac{1}{3}$

2. $y = -\frac{2}{5}x + 1$

$-\frac{2}{5};\ \frac{5}{2}$

3. $y = -x + 4$

$-1;\ 1$

4. $4x + 2y = 10$

$-2;\ \frac{1}{2}$

Type 2 Parallel, Perpendicular or Neither?

Watch Me — Worked Example

$y = 2x + 1$ and $y = -\frac{1}{2}x - 3$: slopes 2 and $-\frac{1}{2}$ multiply to -1. **Answer: perpendicular.**

5. $y = 5x - 2$ and $y = 5x + 9$

parallel

6. $y = \frac{3}{4}x$ and $y = -\frac{4}{3}x + 1$

perpendicular

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7. $y = 2x + 3$ and $y = \frac{1}{2}x - 1$

neither

8. $x + 3y = 6$ and $y = 3x - 2$

perpendicular

Type 3 Write a Parallel Line

Watch Me — Worked Example

Through $(1, -4)$, parallel to $y = -2x + 5$. Same slope -2: $y + 4 = -2(x - 1) \Rightarrow y = -2x - 2$.

9. Through $(3, 1)$, parallel to $y = 2x - 6$.

$y = 2x - 5$

10. Through $(-2, 5)$, parallel to $y = -\frac{1}{2}x + 3$.

$y = -\frac{1}{2}x + 4$

Type 4 Write a Perpendicular Line

Watch Me — Worked Example

Through $(2, 7)$, perpendicular to $y = \frac{2}{3}x - 1$. Slope $-\frac{3}{2}$: $y - 7 = -\frac{3}{2}(x - 2) \Rightarrow y = -\frac{3}{2}x + 10$.

11. Through $(4, -1)$, perpendicular to $y = 2x + 3$.

$y = -\frac{1}{2}x + 1$

12. Through $(-3, 2)$, perpendicular to $y = -3x + 1$.

$y = \frac{1}{3}x + 3$

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13. Through $(0, 5)$, perpendicular to $x - 4y = 8$.

$y = -4x + 5$

14. HARD Find k so that $y = kx+1$ is perpendicular to $3x + 6y = 2$.

$k = 2$

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